

ENGR 315 - Final Project Presentation

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The Pandas Express

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Task 1 - Data of Interest

Data of Interest

We selected a COVID-19 dataset from Data.gov for this project due to its accessibility and analytical potential.

Dataset Access:

- URL: <https://catalog.data.gov/dataset/covid-19-case-surveillance-public-use-data>
-

Data Collection:

- Public health agencies, aggregated by government sources
 - track COVID-19 cases, deaths, and trends over time.

Purpose of the Data:

- monitor the spread of COVID-19
- inform public health decisions
- provide transparency to researchers and the public.

Data Selection Justification

Why This Dataset is Suitable:

- We have prior experience working with its format
- It contains multiple variables that enable comparative and inferential analysis
- It supports time-series analysis and trend identification

Useful Information for Analysis:

- Daily case counts
- Death counts
- Dates (time-series data)
- Geographic location (county/state)

Task 2 - Developing Questions of Interest

Initial Question #1: How does population size influence COVID-19 case rates over time?

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Why this question is interesting:

- Do more densely populated areas experience a faster/more severe spread of COVID-19?

Why this is an inferential question:

- Explores the relationship between population and case rates
- Requires analysis over time to identify trends

Variables of interest:

- Population size
- Daily COVID-19 case counts
- Time (date)

Initial Question #2: Is there a temporal lag relationship between new COVID-19 cases and subsequent deaths?

Why this question is interesting:

- Helps better understand disease progression
- Improves predictions for healthcare demand

Why this is an inferential question:

- Analyzing time-shifted data
- Determine whether one variable (cases) influences another (death) after a delay

Variables of interest:

- Daily case counts at time t
- Daily death counts at time $t + \text{lag}$
- Time (date)

Initial Question #3: How do peak infection rates differ between rural and urban regions?

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Why this question is interesting:

- Comparing rural and urban areas
- Highlights how geographic/demographic differences affect the spread/severity of COVID-19

Why this is an inferential question:

- Compares groups
- Identifies patterns/differences in peak values

Variables of interest:

- Region classification (rural vs. urban)
- Maximum (peak) case counts
- Location (county/state)

Initial Question #4: What trends can be identified in COVID-19 case spikes over time (e.g., waves)?¹⁰

Why this question is interesting:

- Analyzing trends and waves helps understand how the pandemic evolved and whether patterns repeat over time.

Why this is an inferential question:

- Interpreting patterns in time-series data
- Identifying meaningful trends, rather than answering a yes/no question.

Variables of interest:

- Time (date)
- Daily case counts

Initial Question #5: Does a higher initial outbreak correspond to a faster rate of decline in cases?

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Why this question is interesting:

- Explores whether regions w/ severe early outbreaks recover more quickly
- Provides insight into immunity/intervention effects

Why this is an inferential question:

- Examines the relationship between two calculated metrics (initial spike and rate of decline),
- Requires analysis and interpretation

Variables of interest:

- Initial peak case count
- Rate of decrease in cases over time
- Time (date)

Initial Question #6: How does applying a 7-day moving average affect the interpretation of COVID-19 trends?

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Why this question is interesting:

- Smoothing data can significantly change how trends appear
- Impacts how conclusions are drawn from the data.

Why this is an inferential question:

- Compares two representations of the same data to evaluate how data processing affects interpretation.

Variables of interest:

- Raw daily case counts
- 7-day moving average of cases
- Time (date)

Task 3 - Data/Question Alignment

Data/Question Alignment

Question	Dataset Variables Used	Can it Answer Question?
How does population size influence COVID-19 case rates over time?	Population data (external or merged), daily case counts, date	Maybe (No)
Is there a temporal lag relationship between new COVID-19 cases and subsequent deaths?	Daily case counts, daily death counts, date	Yes
How do peak infection rates differ between rural and urban regions?	Case counts, county/state location (urban/rural classification may require external data)	Maybe (No)
What trends can be identified in COVID-19 case spikes over time (e.g., waves)?	Daily case counts, date	Yes
Does a higher initial outbreak correspond to a faster rate of decline in cases?	Case counts, date (used to calculate peak and rate of decline)	Yes
How does applying a 7-day moving average affect the interpretation of COVID-19 trends?	Daily case counts, date (used to compute moving average)	Yes

Feedback From Dr. Morgul

- Questions were overly simple
- Did not tell a story
- Similarity between Q1 and Q3
- Needed a more complex question

Final Question #1: Is there a temporal lag relationship between new COVID-19 cases and subsequent deaths?

Justification/Rough Plot Idea

- Converted cumulative → daily data
- Tested lag values (0–30 days)
- Shifted deaths to align with cases
- Calculated correlation at each lag
- Selected lag with highest correlation

Axes

- X-axis: Time (days)
- Left Y-axis: Daily cases
- Right Y-axis: Daily deaths
- Lag plot: Cases vs shifted deaths

Expected Result

- Optimal lag time: 2–3 weeks
- Weak correlation

Final Question #2: How does the moving average size affect trend interpretation?

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Justification/Rough Plot Idea

- Converted cumulative daily data
- Applied 3, 7, 14, and 30-day moving averages
- Compared smoothed trends to raw data
- Counted peaks at each window size
- Zoomed into Wave 4 for detail

Axes

- X-axis: Time (days)
- Y-axis: Daily cases
- Peak plot X-axis: Window size (days)
- Peak plot Y-axis: Number of detected peaks

Expected Result

- Larger windows smooth data more
- Peak count decreases as window size increases
- Larger windows reduce short-term variation

Final Question #3: How does growth rate differ between outbreak waves?

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Justification/Rough Plot Idea

- Converted cumulative daily data
- Split data into 4 outbreak waves
- Found peak daily cases in each wave
- Calculated days to peak
- Calculated growth rate per wave

Axes

- X-axis: Days to peak
- Y-axis: Peak daily cases
- Guide lines: Peak and time references

Expected Result

- Wave 4 had the highest peak and fastest growth
- Earlier waves grew slower and peaked lower
- Later waves showed stronger outbreak growth

Task 4 - Metrics, Tools, & Plots

Question 1

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Metrics

- Lag Time
- Peak alignment

Additional Python Packages

- pandas (data handling, time shifting)
- numpy (correlation calculations)
- matplotlib (visualization)

Plots/Visualization

- Overlaid time series

Question 2

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Metrics

- Standard deviation
- Peak count

Additional Python Packages

- pandas (data handling, time shifting)
- numpy (correlation calculations)
- matplotlib (visualization)

Plots/Visualization

- Multiple smoothed curves
- Peak count comparison

Question 3

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Metrics

- Slope
- Peak height
- Time to peak

Additional Python Packages

- pandas (data handling, time shifting)
- numpy (correlation calculations)
- matplotlib (visualization)

Plots/Visualization

- Wave comparison bar graph
- Case curves

Task 5 - Sketches at Answers

Question 1: Calculated Metrics

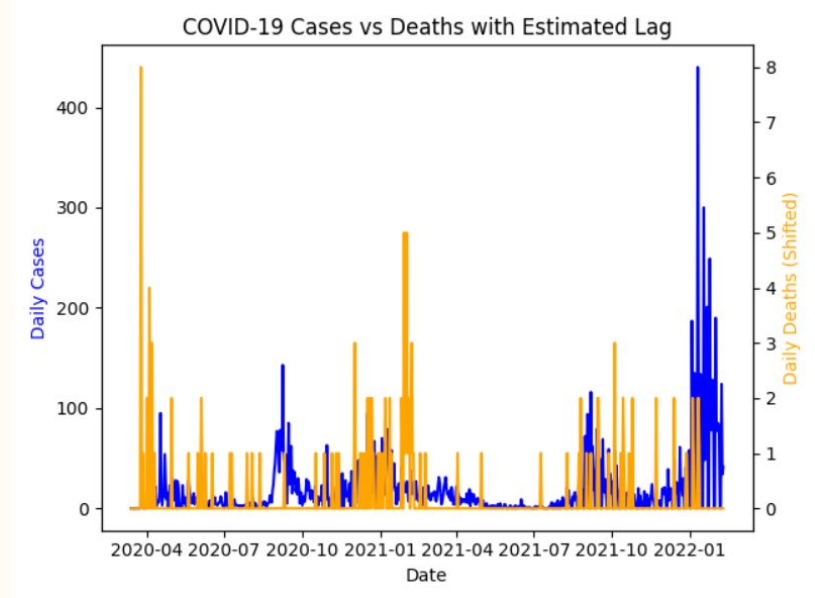
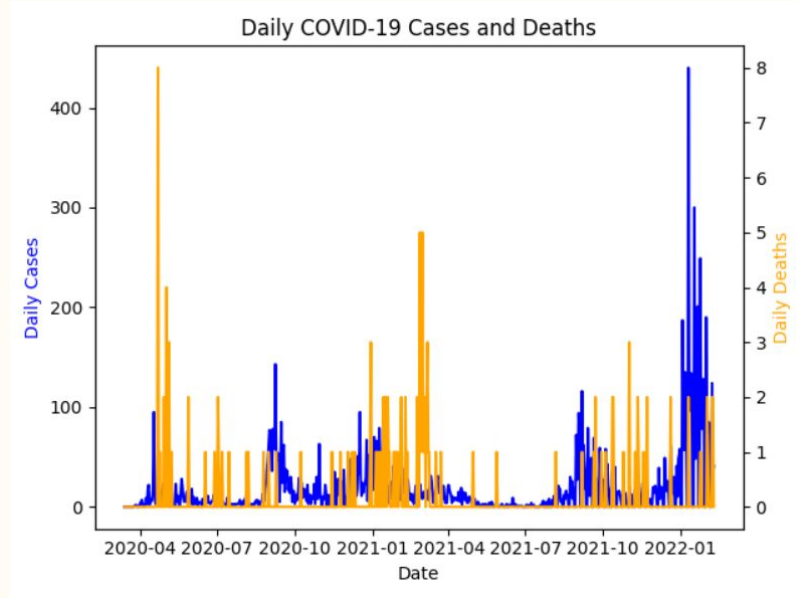
QUESTION 1: Temporal Lag Between COVID-19 Cases and Deaths

Location: Harrisonburg city, Virginia

Best estimated lag: 28 days

Correlation at best lag: 0.154

Question 1: Plots



Question 2: Calculated Metrics

Question 2: Moving Average Window Comparison

Raw Daily Cases

Standard deviation: 33.61023839264073

3-Day Moving Average

Standard deviation: 24.916391497070347

Number of detected peaks: 55

7-Day Moving Average

Standard deviation: 22.85752532151746

Number of detected peaks: 37

14-Day Moving Average

Standard deviation: 21.838964390126268

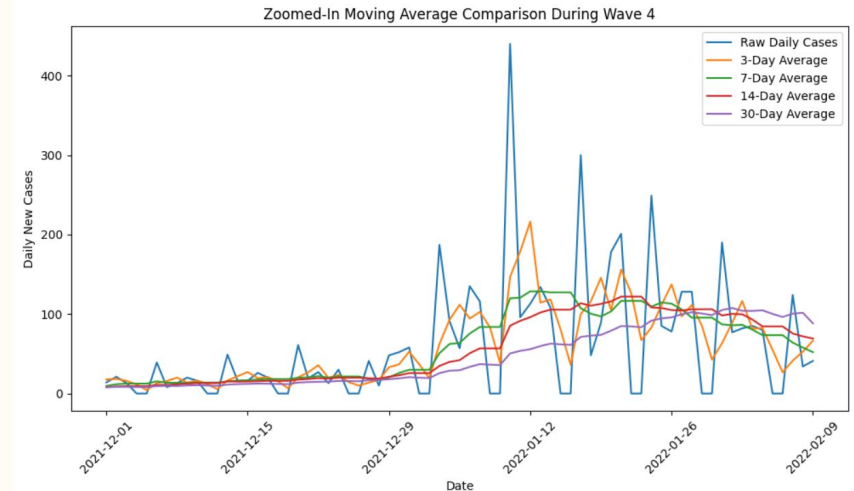
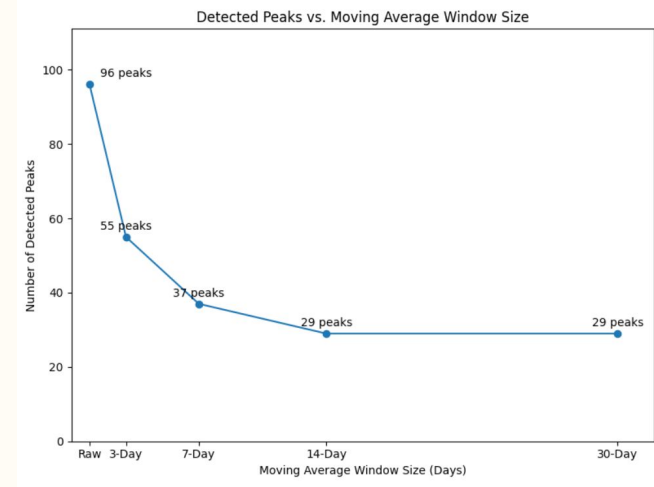
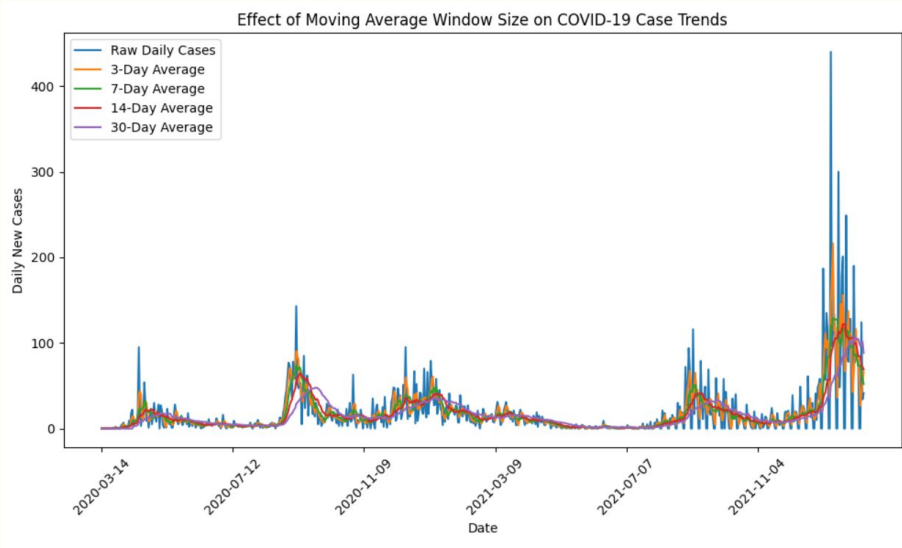
Number of detected peaks: 29

30-Day Moving Average

Standard deviation: 18.62355702054612

Number of detected peaks: 29

Question 2: Plots



Question 3: Calculated Metrics

Wave 1

Date range: 2020-03-01 to 2020-06-30

Peak daily cases: 95

Peak date: 2020-04-17

Days from wave start to peak: 35

Average growth rate: 2.7142857142857144 cases per day

Wave 2

Date range: 2020-10-01 to 2021-02-28

Peak daily cases: 95

Peak date: 2020-12-17

Days from wave start to peak: 78

Average growth rate: 1.2179487179487178 cases per day

Wave 3

Date range: 2021-07-01 to 2021-10-31

Peak daily cases: 116

Peak date: 2021-09-06

Days from wave start to peak: 68

Average growth rate: 1.7058823529411764 cases per day

Wave 4

Date range: 2021-12-01 to 2022-02-28

Peak daily cases: 440

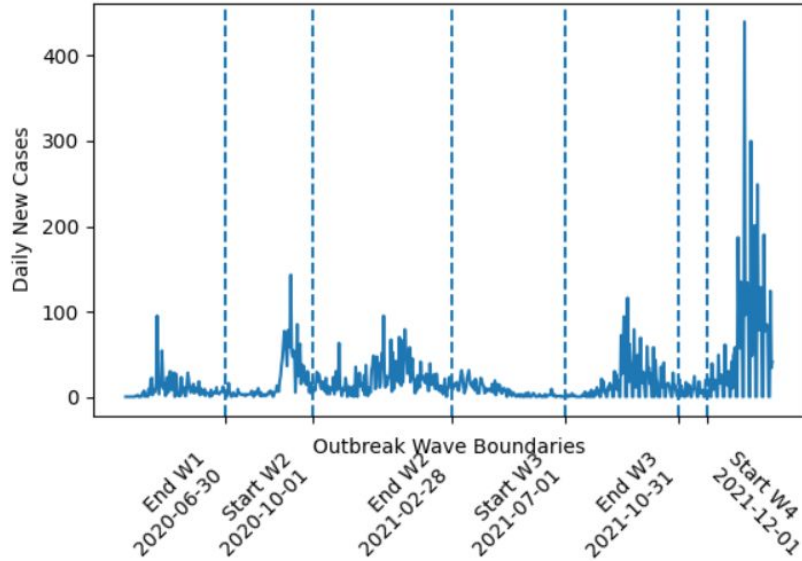
Peak date: 2022-01-10

Days from wave start to peak: 41

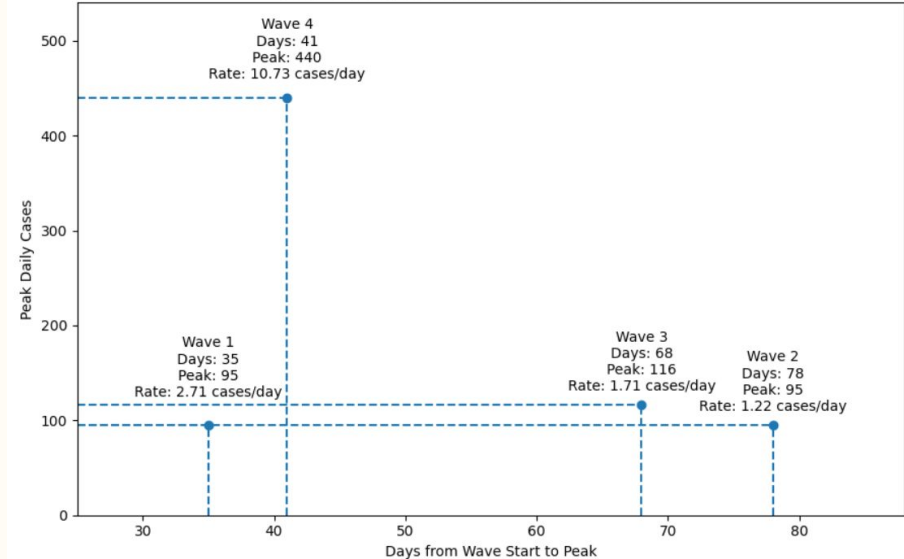
Average growth rate: 10.731707317073171 cases per day

Question 3: Plots

Daily COVID-19 Cases Over Time with Outbreak Wave Boundaries



Peak Severity vs. Time to Peak by Outbreak Wave



Task 6 - Data Decisions

Question #1: Temporal Lag

What answers did you find to the original questions?

- The temporal lag is about 28 days between cases and deaths when testing 0-30 days of lag

Did you find the relationships among the variables that you expected?

- We expected a temporal lag around 3 weeks to be optimal

Can a clear conclusion or trend be seen

- Results showed a lag of approximately 28 days (4 weeks) with a weak correlation (0.154)
- The weak relationship is likely due to low and variable death counts at the county level, does not take into account health characteristics of infected individuals

Question #2: Moving Average

What answers did you find to the original questions?

- The detected peaks trended towards 29 peaks as the moving average time moved
- Standard deviation decreased

Did you find the relationships among the variables that you expected?

- We expected the standard deviation to decrease
- Did not expect the trend towards 29 peaks

Can a clear conclusion or trend be seen

- Smoothing data and reducing noise makes important statistical data easier to work with

Question #3: Growth Rate For Different Waves

What answers did you find to the original questions?

- Wave 4 (2021-12-01 to 2022-02-28) had the highest peak daily cases and growth rate

Did you find the relationships among the variables that you expected?

- Yes, we expected the Christmas/New Years time period to have the highest daily cases/growth rate
- Unexpected difference between Wave 2 (2020) and Wave 4 (2021)

Can a clear conclusion or trend be seen

- The winter holiday season is where a lot of people travel to visit friends/families, which most likely resulted in a higher spread of COVID-19

Task 7 - Reflection

Project Reflection

- Many interesting analysis techniques to analyze COVID data
 - Correlation, scatter plots, bar graphs
- Confirmed trends between holiday season and the spread of COVID
- What would we do differently?
 - Our results matched what we expected very well, we would try to branch out more
 - We would have taken into account different demographics and how they are impacted by COVID.

Class Reflection

- Learned a lot about different statistical analysis techniques
- Gained foundational knowledge of coding in python, which helped in both capstone and other classes
- Learned to troubleshoot code and understand error messages



Questions?